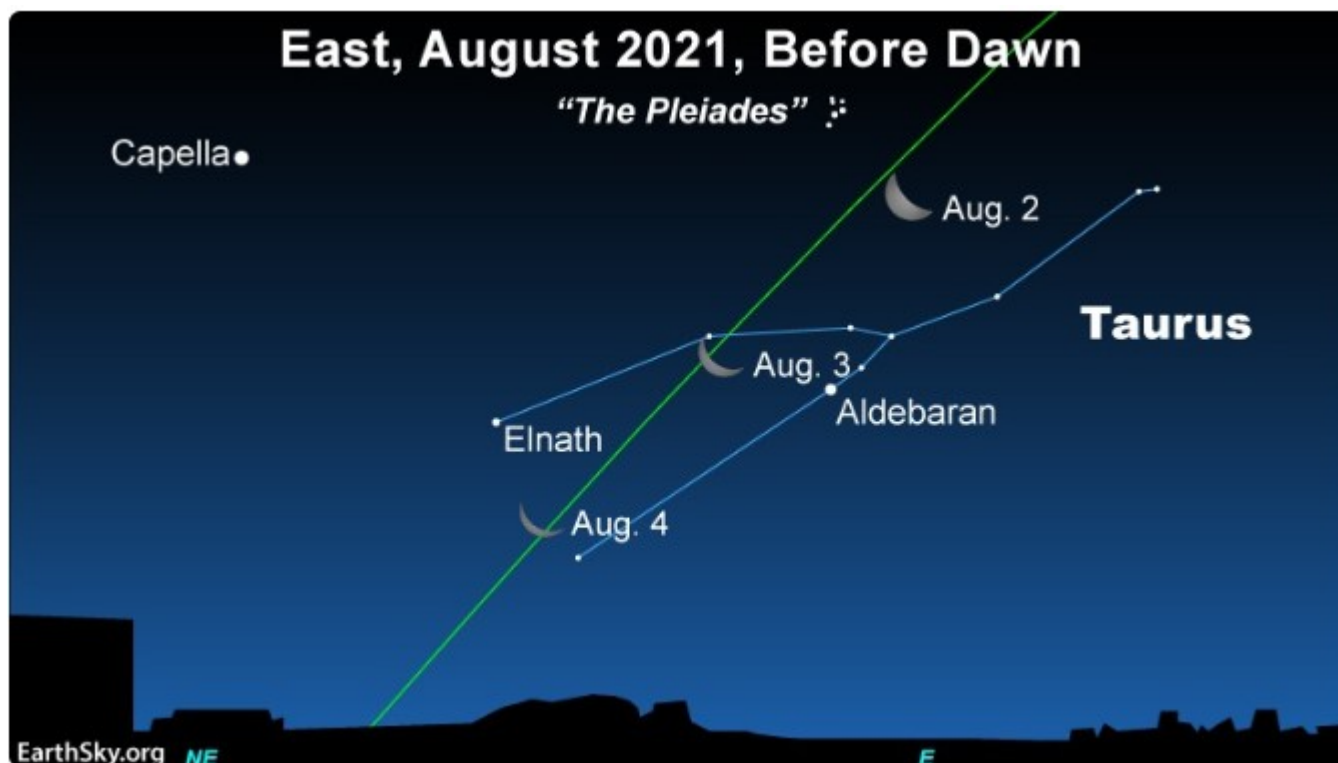




The sky through BINOCULARS And simple TELESCOPES

<https://earthsky.org/tonight/>

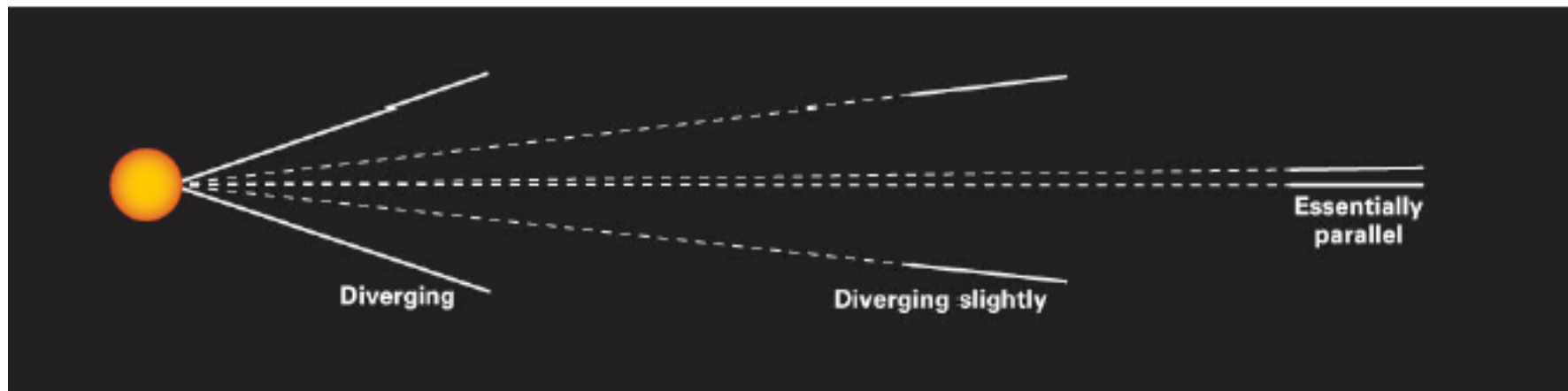


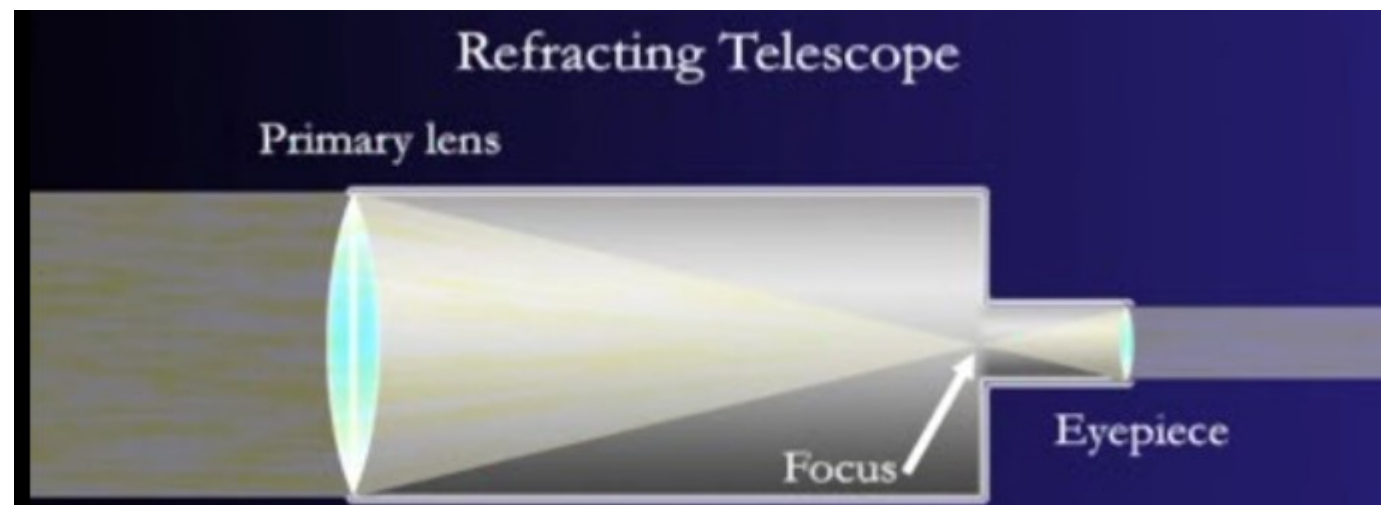
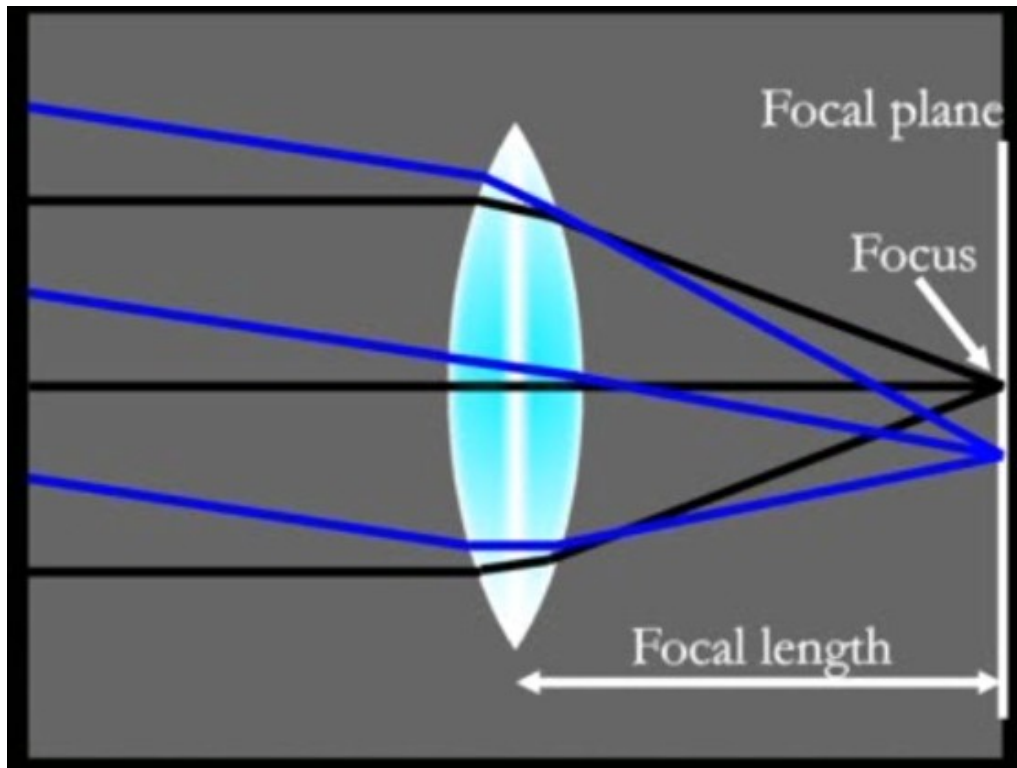
<https://www.youtube.com/watch?v=EieqUGAJHMw>

**Rays from stars hit lenses from binoculars and telescopes
In a // fashion**

Light rays from a star are essentially parallel

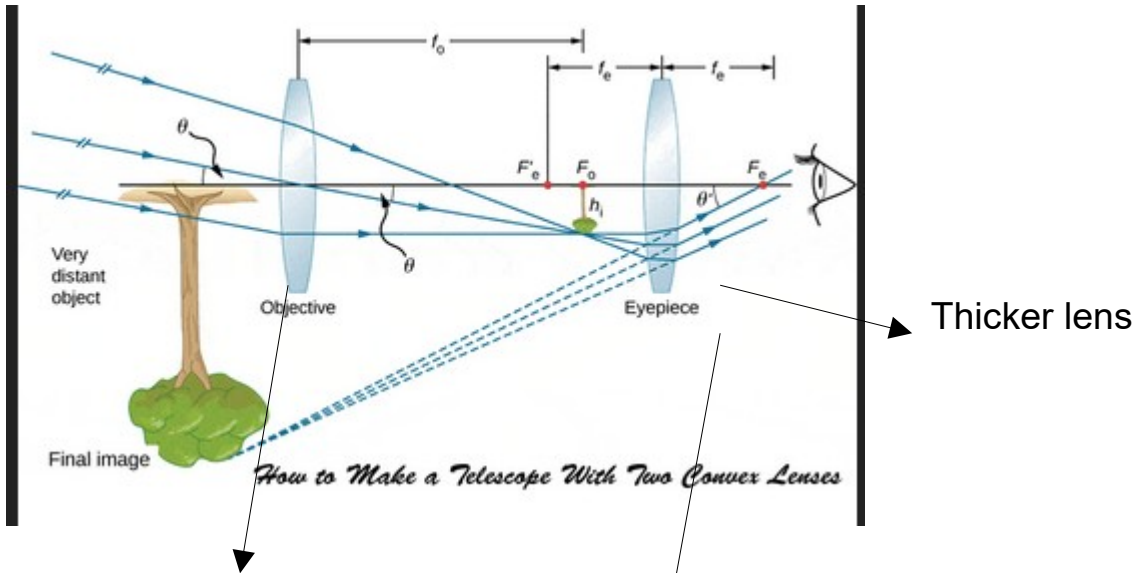
Stars are very, very far away, so the rays reaching us diverge only very slightly. They are PARALLEL for all practical purposes.



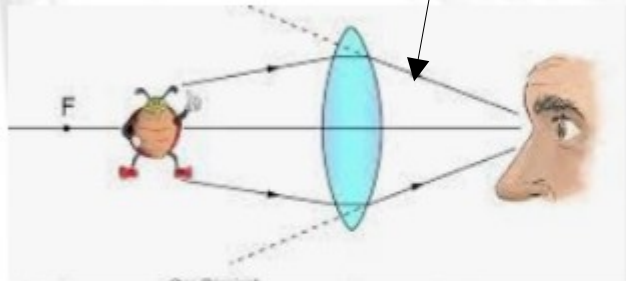


See animation telescope

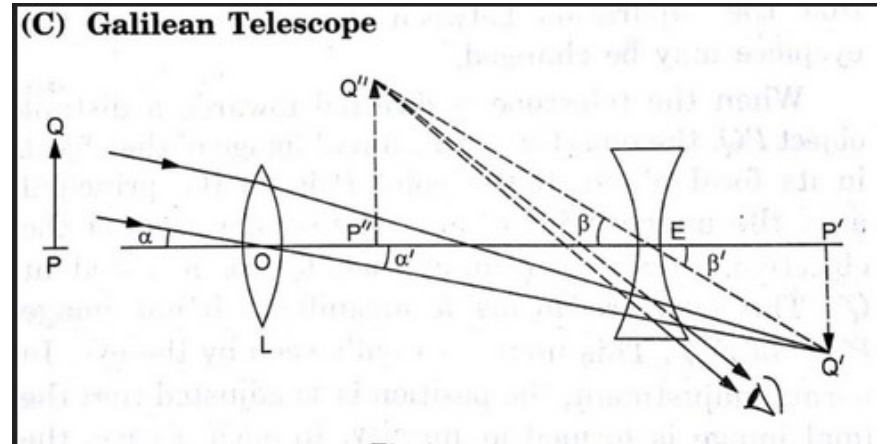
simulation



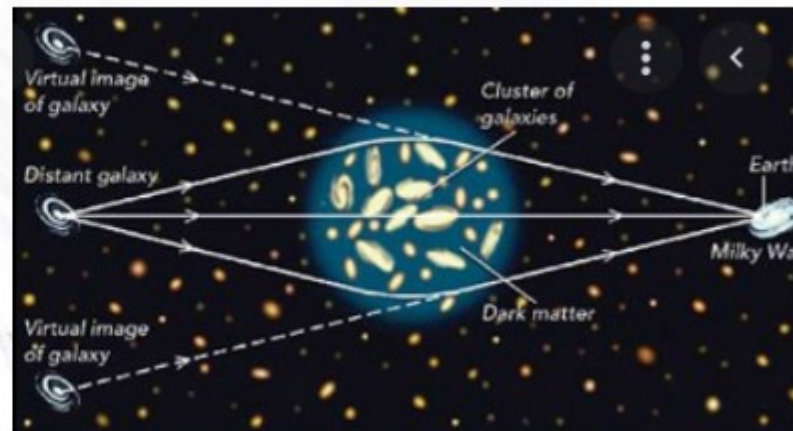
objective → image is formed at the focus



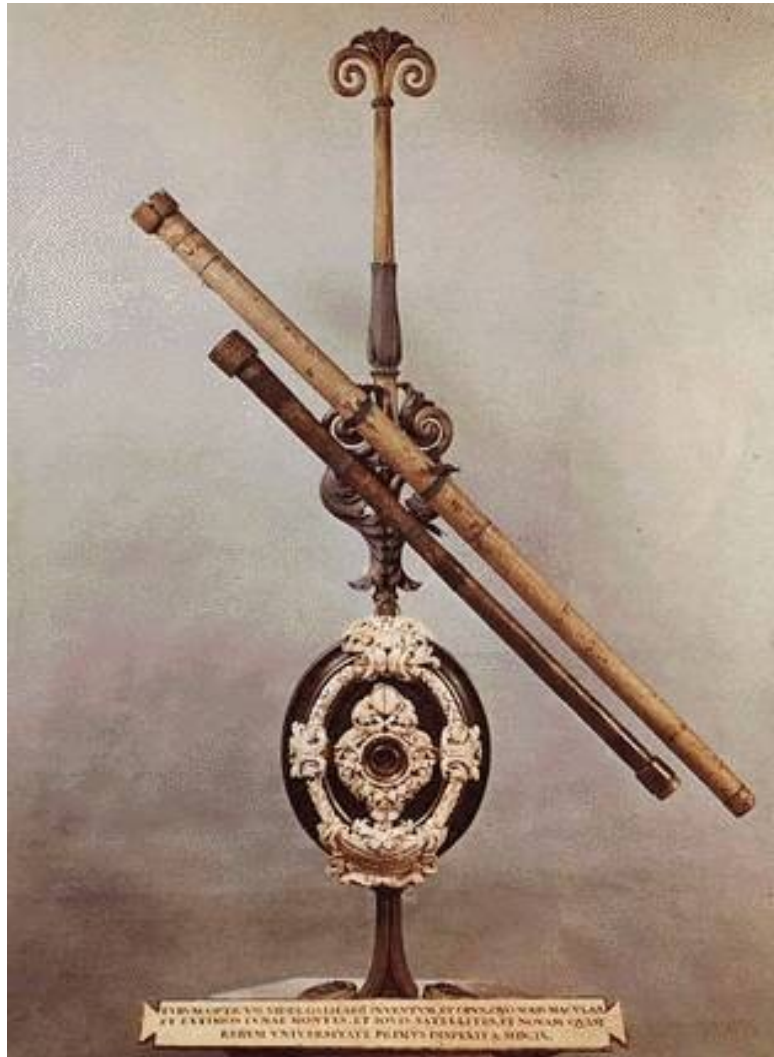
Eye piece → magnifying glass



Gravitational lensing



Galileo refracting telescope Florence's science museum.



Galileo discovered:

- moons of Jupiter
- phases of Venus
- sunspots
- mountains and valleys on the moon
- ring of Saturn

Changed our place in the Universe.
Change our view of the Universe.

Breaking discoveries in Astronomy

Hans Lippershey (c1570-c1619) of Holland is often credited with the invention (was a toy) that Galileo improved. He was a Deutch optician.

Binoculars = pair of refractive telescope.

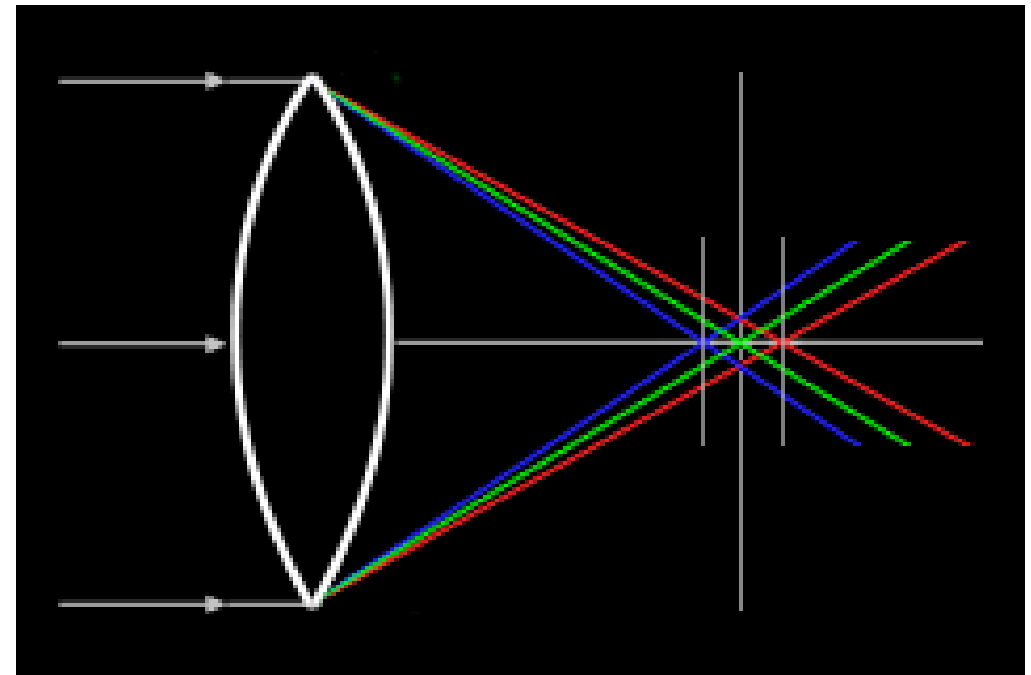
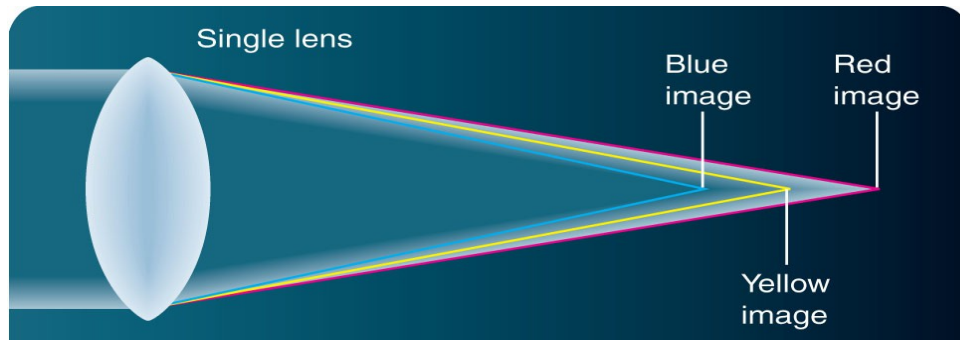


Problems with telescopes:

- Chromatic aberration

optic phenomenon due to the Bending of light as a function of color is a problem with Refractive telescope.

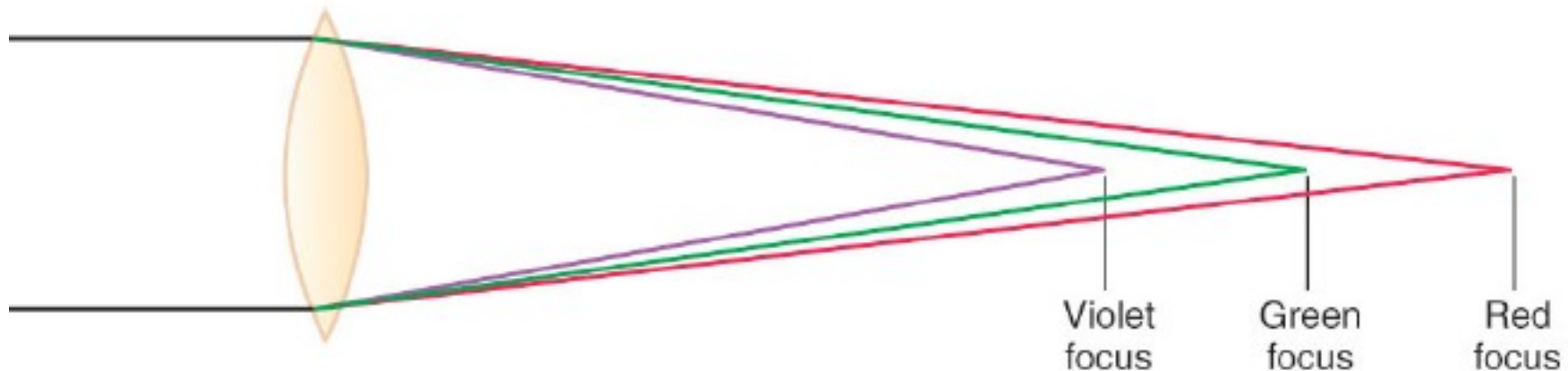
Solution = reflective Telescope.



Physics behind it

Refracting Telescopes (cont.)

- Simple to make, but they suffer from several problems, including **chromatic aberration**
- This is a property of all lenses...including your eyes!



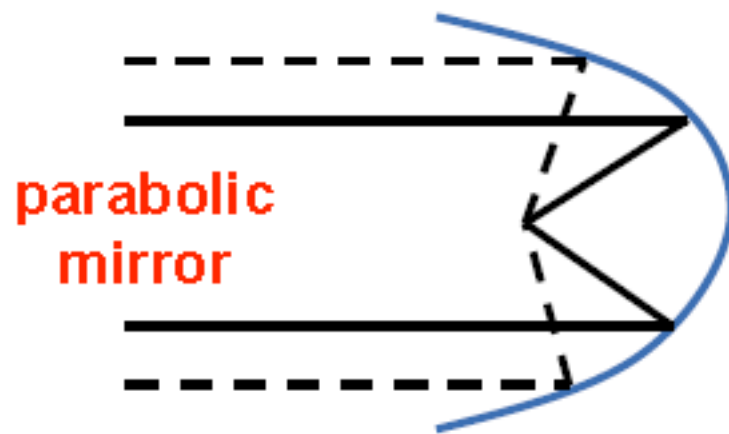
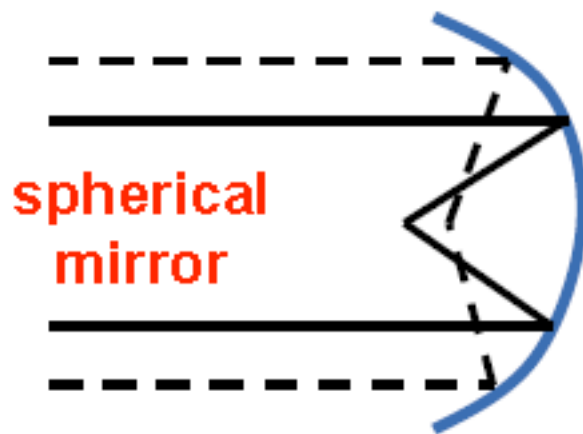
© 2007 Thomson Higher Education

- Lenses are also heavy and absorb light

You can fix that by having layers of different size lenses. To force light to bend the same Way. But too hard for heavy lenses and large ones.

Reflecting Telescopes

- Use a **mirror** to bring light to a focus.

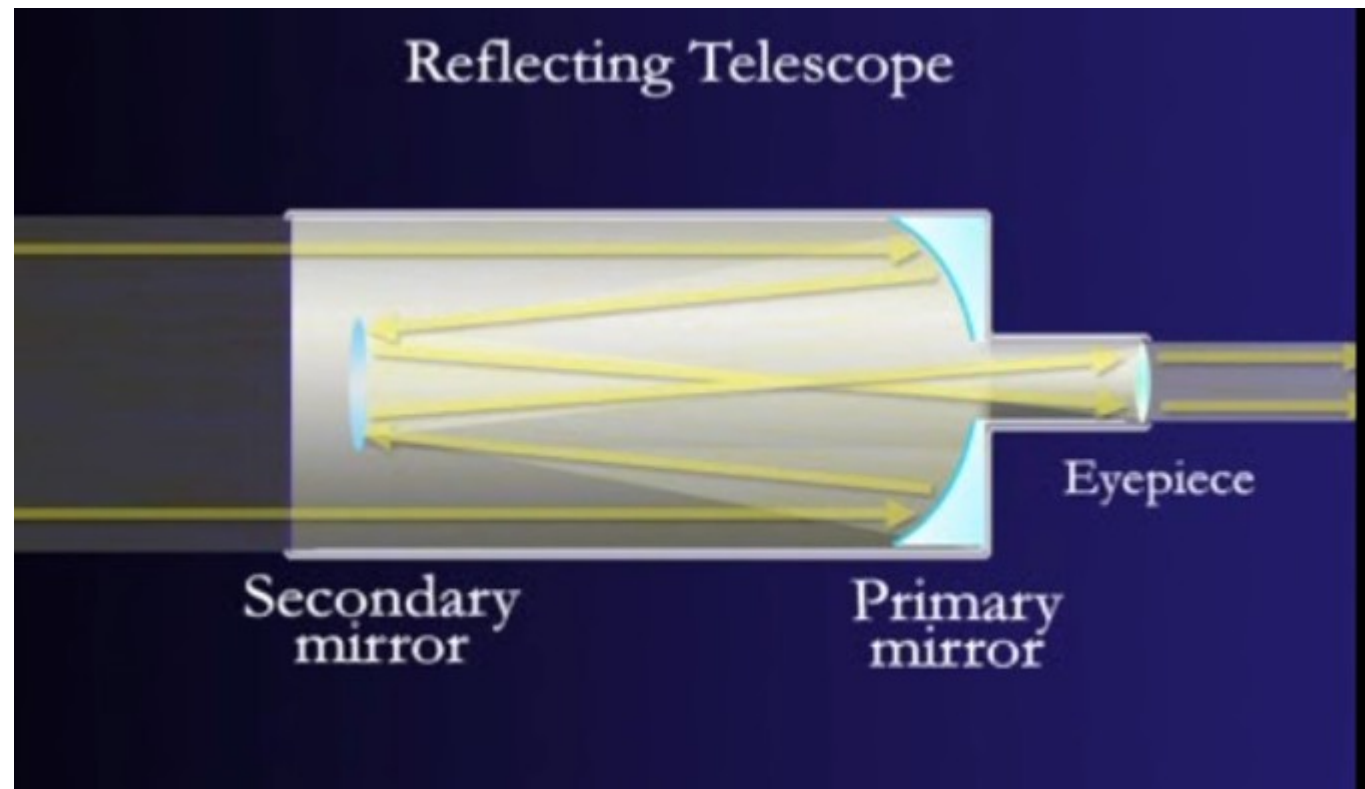
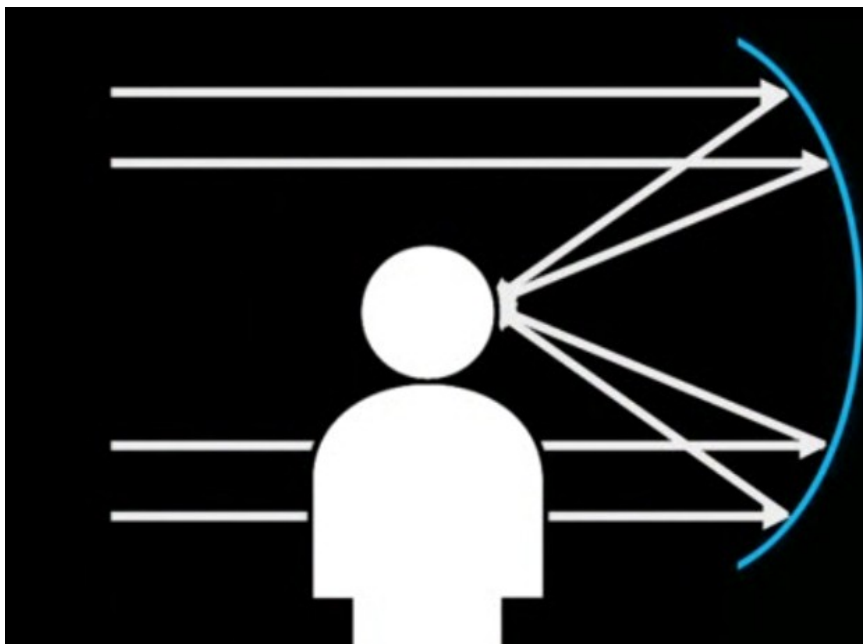


- A spherical mirror does not focus the light to a single point, which leads to **spherical aberration**.
- A parabolic mirror does focus the light to a single point.
- This was in fact the problem that Hubble first had.

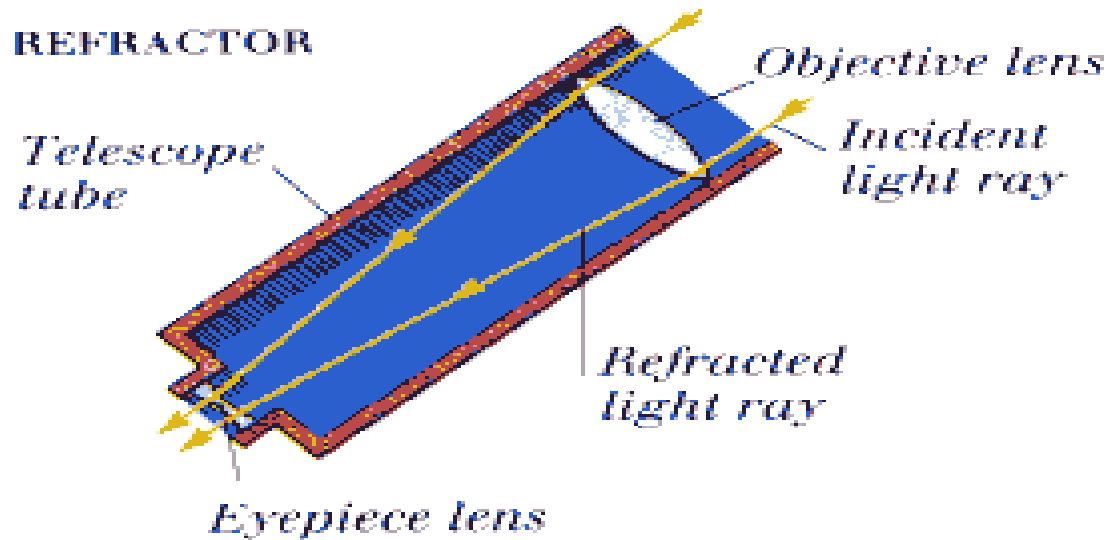


The first reflective
Telescope was made
By Isaac Newton and
Donated to the Royal
Society of London.
1660. this is How Newton
Became a member of
The society.
(replica by F L Agate
1924)





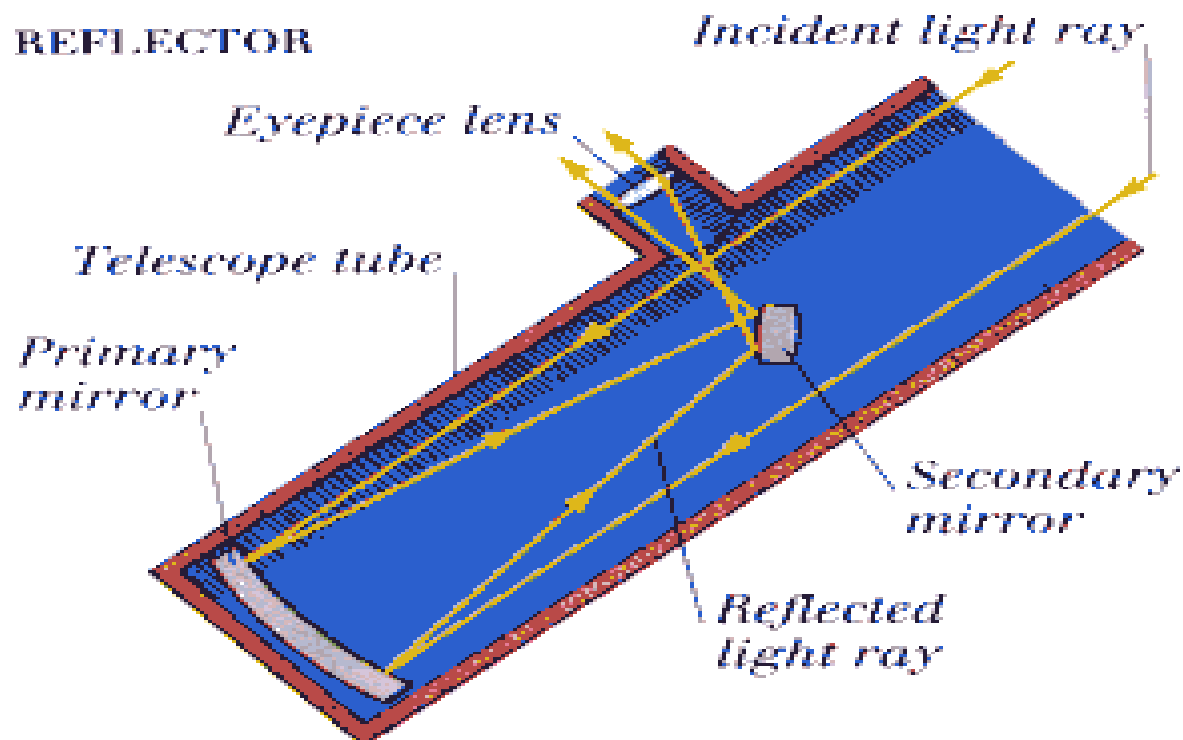
REFRACTOR



Big refractive telescope
Needs a large lens at the top.

It's hard to design a large
Good quality lens and gravity
Deforms the shape.

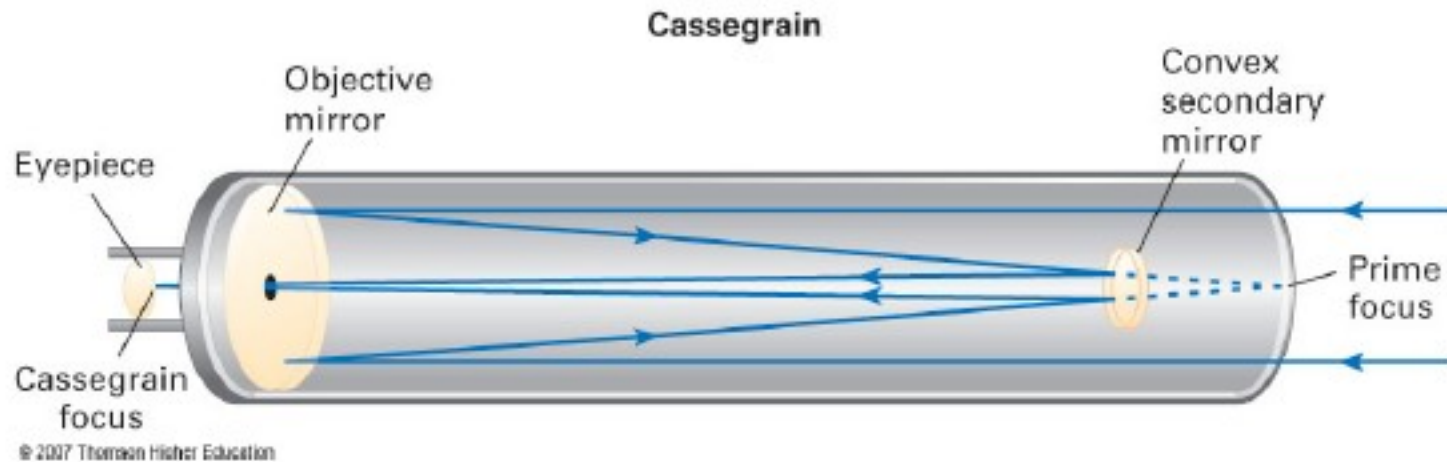
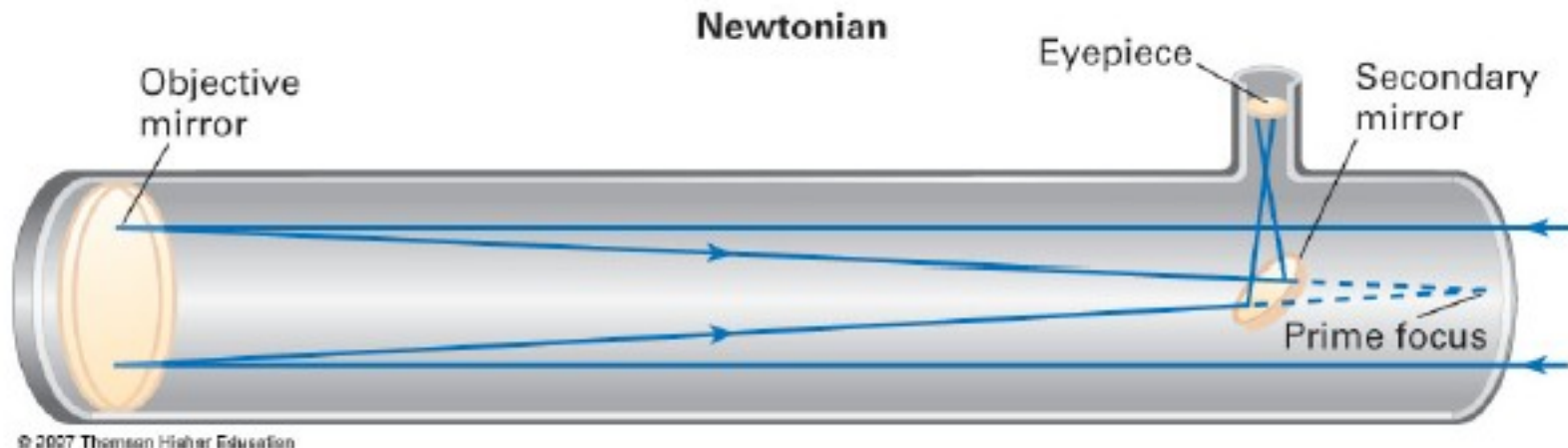
REFLECTOR



A mirror in a reflective telescope
Is place at the bottom so
Gravity is less a problem.

Only the cover of the mirror
Has to be of good quality.
Not the inside glass.

Common Reflector Designs



We need a second mirror so the instrument does not block the rays.
The Newtonian was invented by Newton

2



33% OFF

Celestron StarSense Explorer DX 102AZ Smartphone App-Enabled Refractor Telescope

SKU: CNSSEDX102 MFR: 22460

★★★★★ (44) reviews 8 Questions, 108 Answers

\$399.95 You save: **\$200.00 (33%)** OR **\$67/mo** suggested payments with 6-month financing. [Learn how.](#)

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1

Add to Cart

[Add to wish list](#)
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20mm Erecting Eyepiece Power Newtonian Reflective Astronomical Telescope Space

Condition: New

Bulk savings:

Buy 1

\$40.16/ea

Buy 2

\$38.15/ea

Quantity:

1

2 available / [1 sold](#)

Price: **US \$40.16/ea**

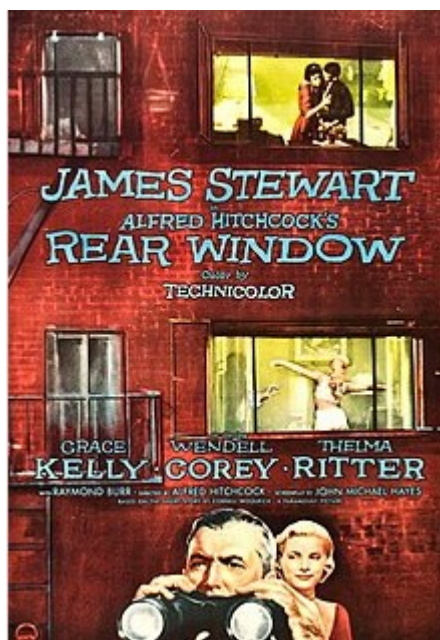
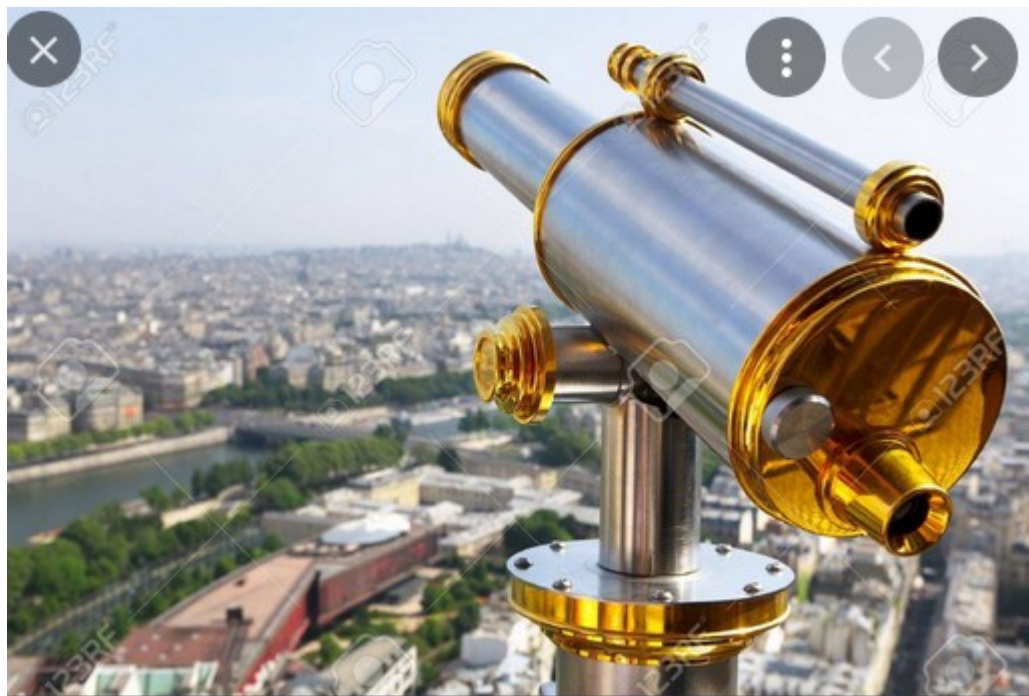
Buy It Now

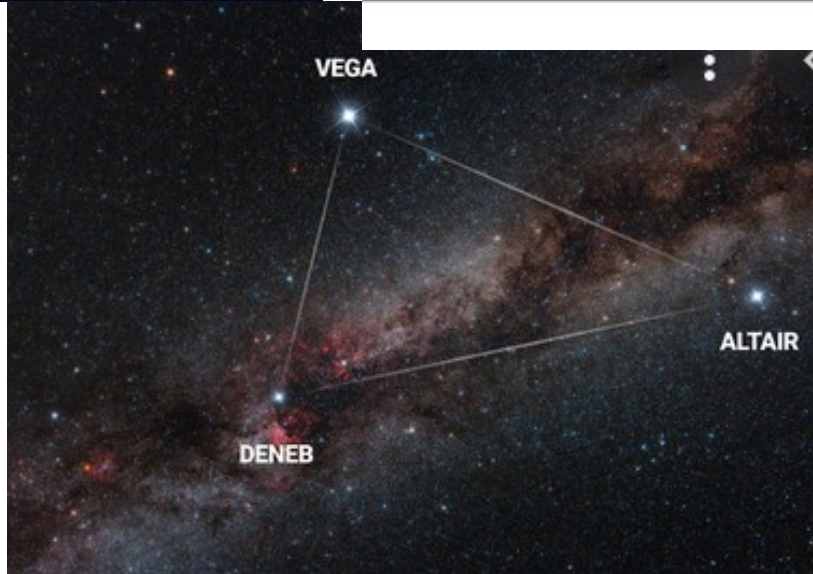
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Best Offer:

Make Offer

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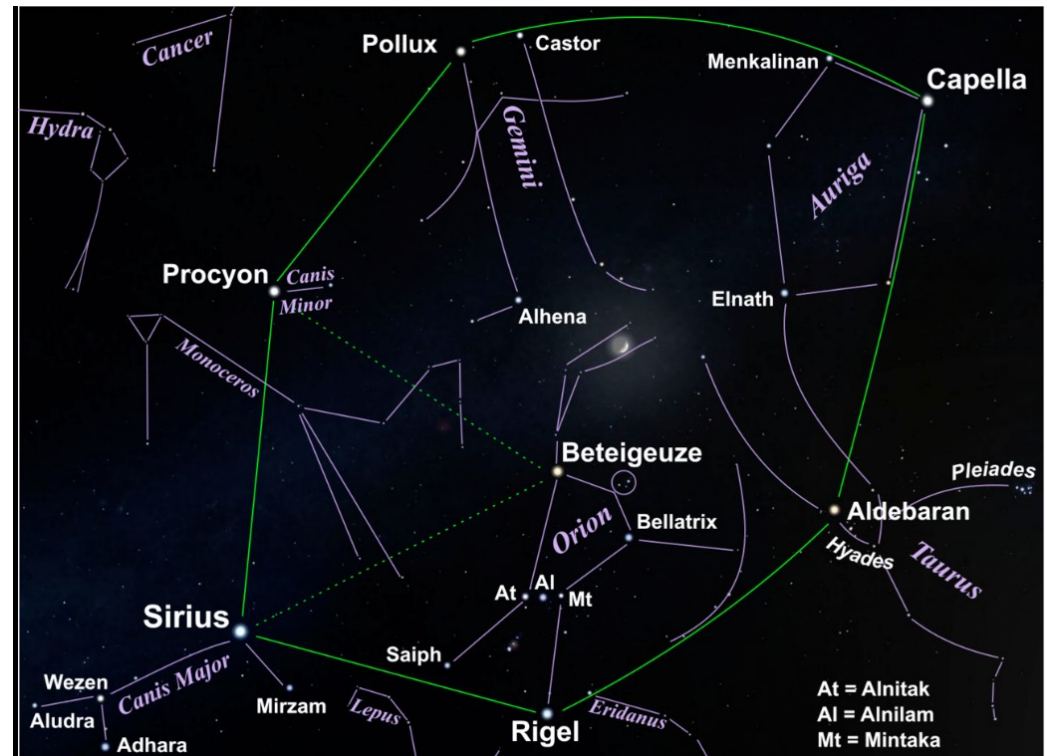




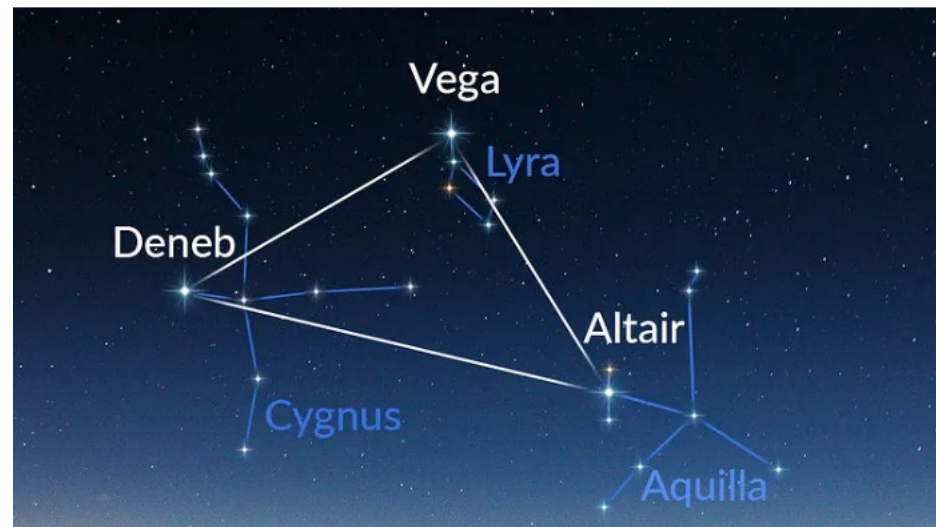
See video.
And next slide

The Summer Triangle is an astronomical asterism in the northern celestial hemisphere. The defining vertices of this imaginary triangle are at Altair, Deneb, and Vega, each of which is the brightest star of its constellation (Aquila, Cygnus, and Lyra, respectively).

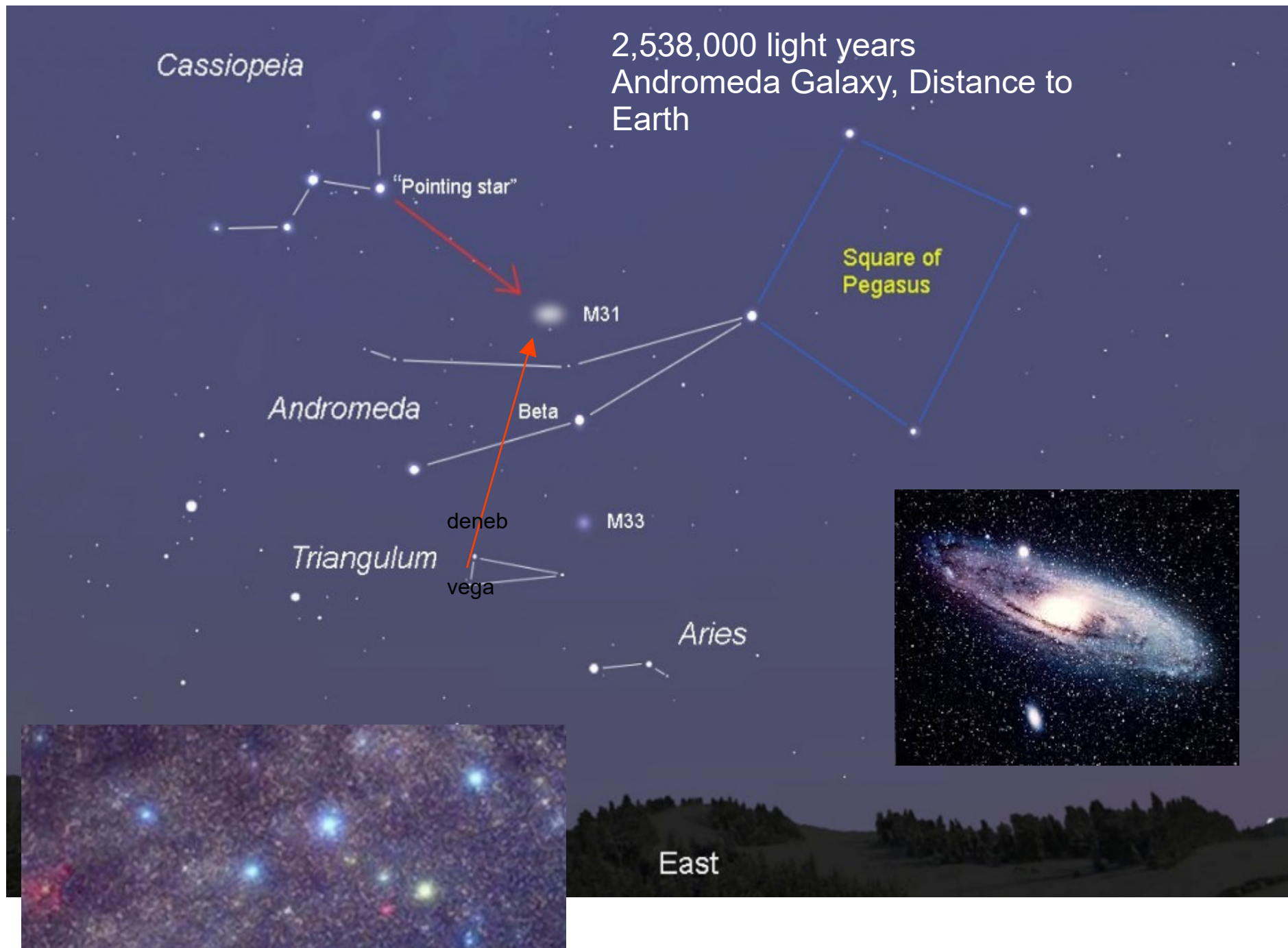
The Winter Hexagon or Winter Circle/Oval is an asterism appearing to be in the form of a hexagon with vertices at Rigel, Aldebaran, Capella, Pollux, Procyon, and Sirius. video



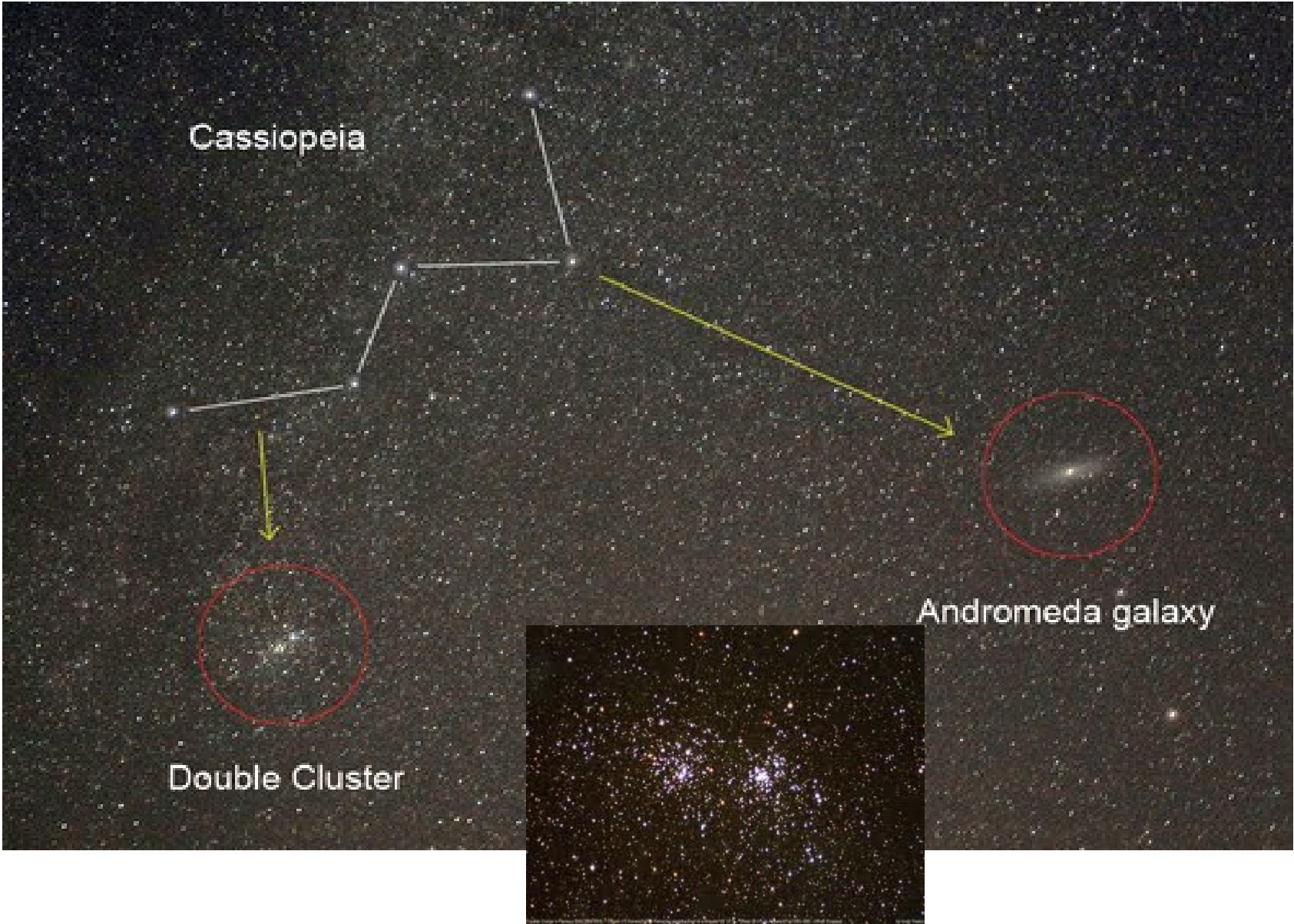
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Andromeda galaxy can be seen from Triangulum or Cassiopeia

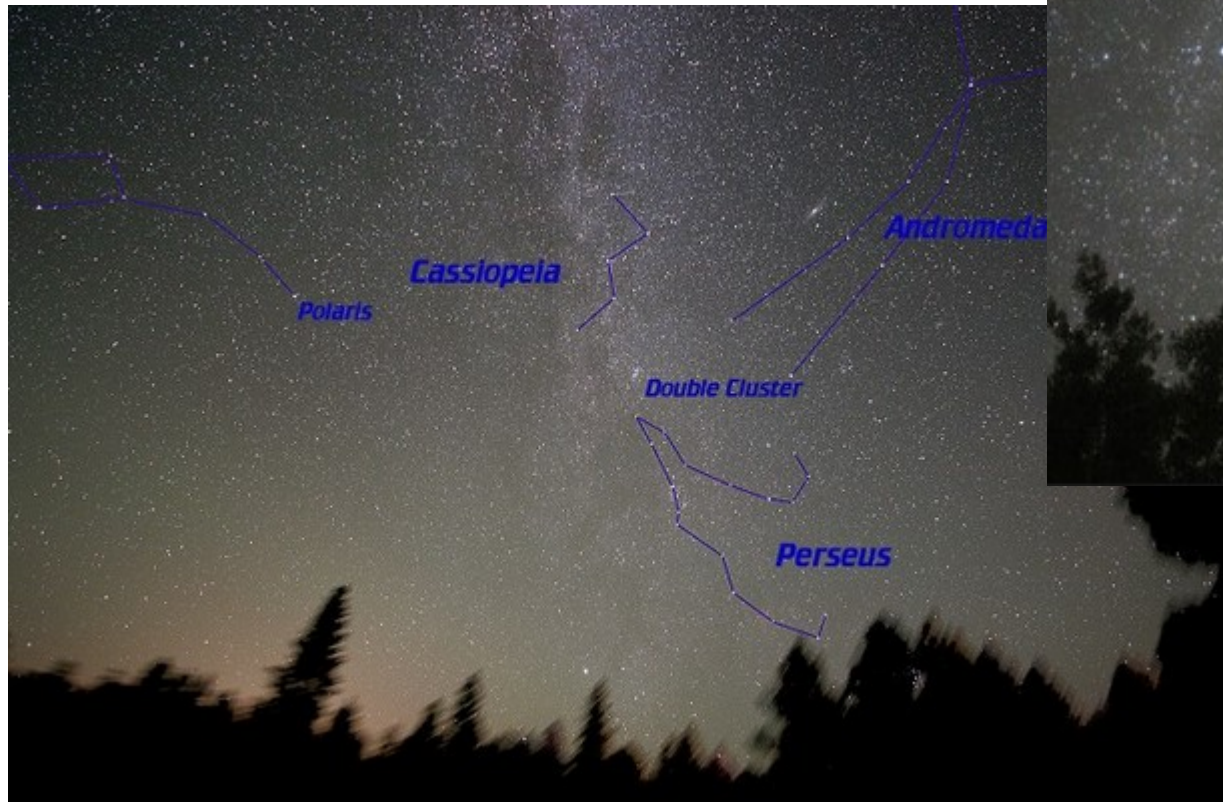


Between the W of Cassiopeia and Perseus you can look at the pair of open cluster called double clusters in Perseus.

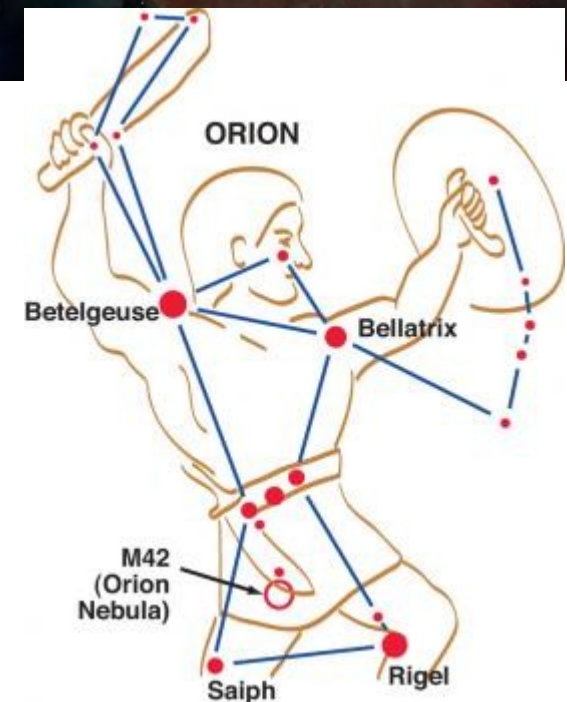


Double cluster in perseus





Orion in the sword, Orion nebula. few millions light years.
radiation illuminate the cloud of stars. new forming stars

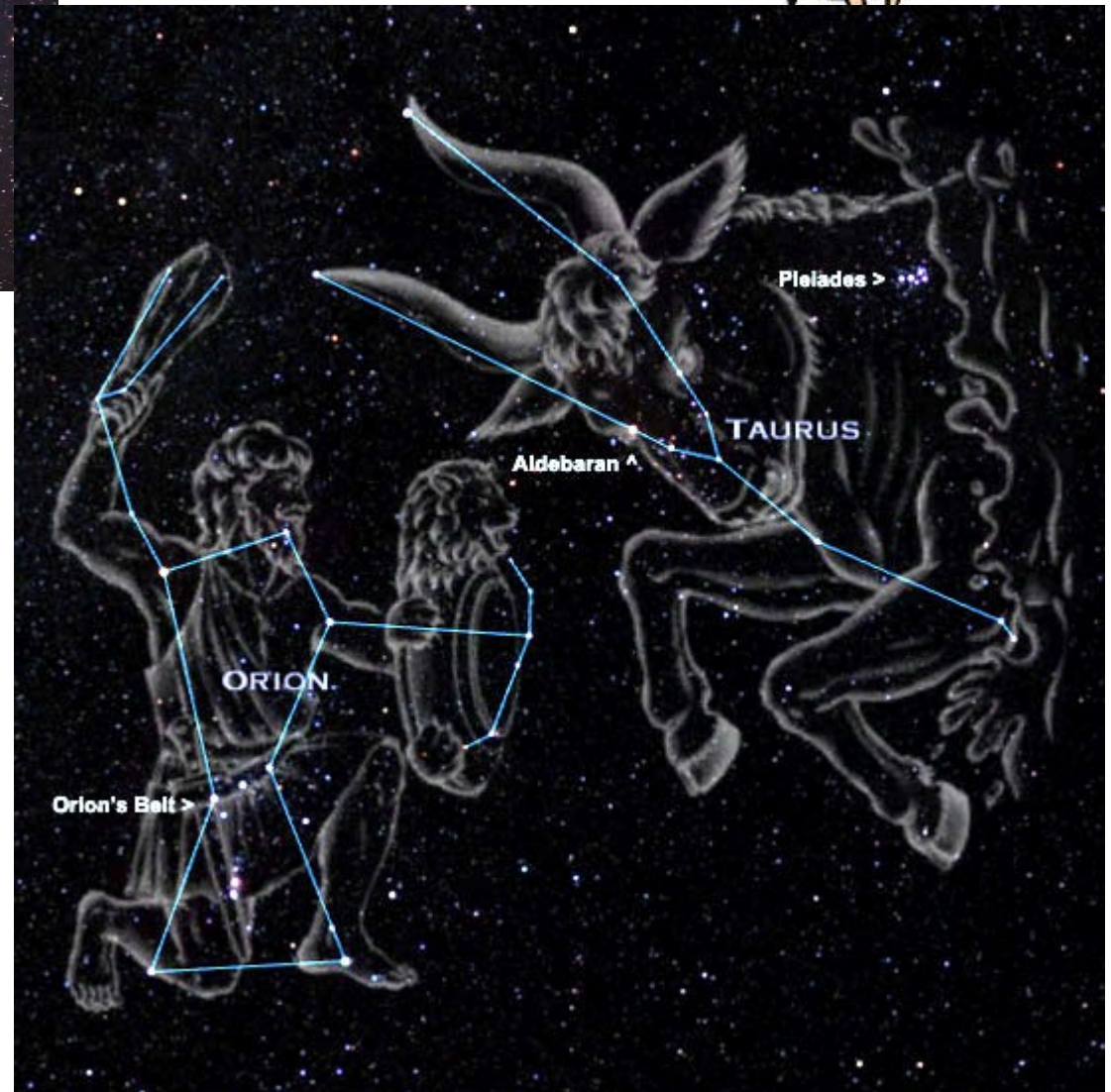




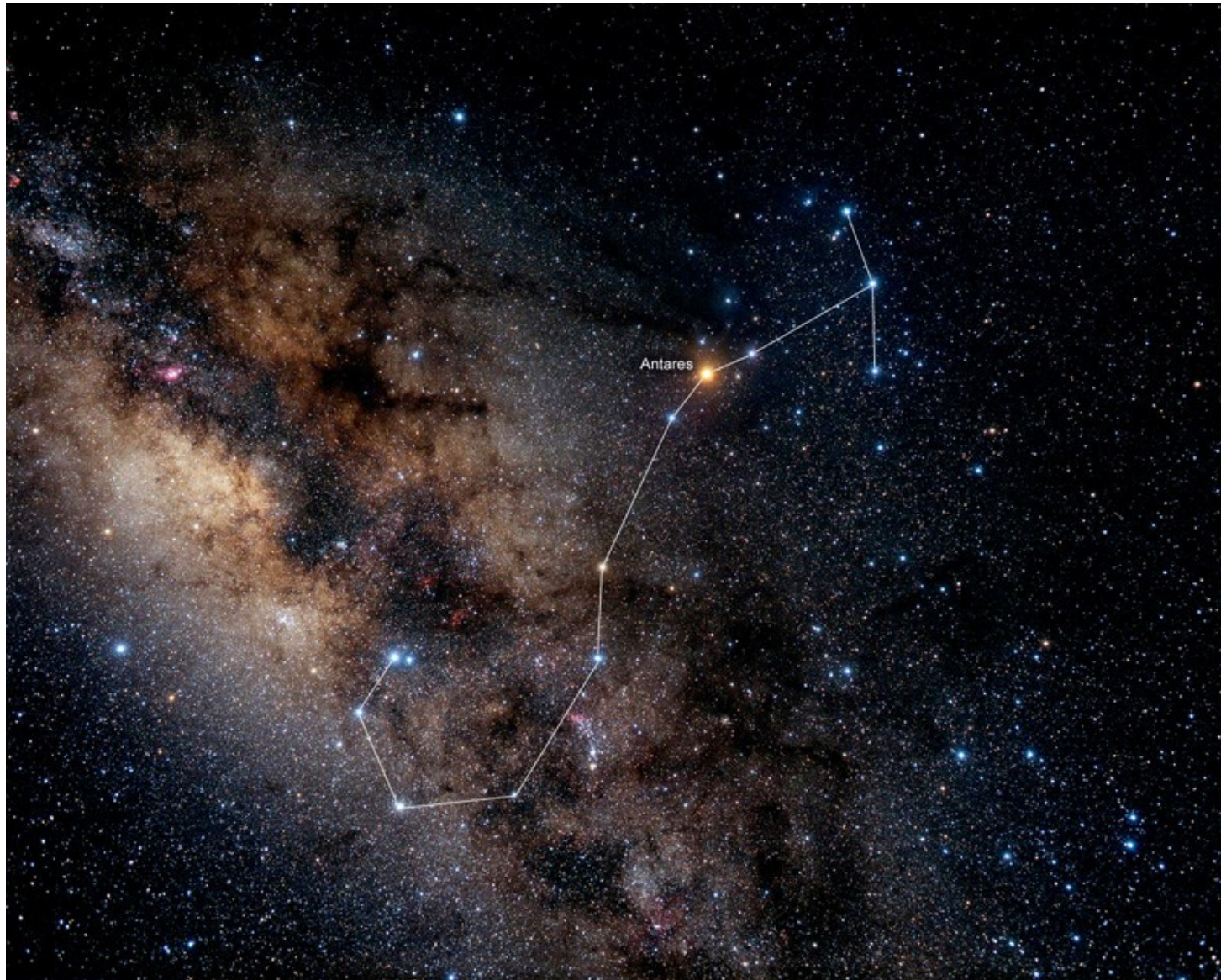




pleiades (cluster) or 6 sisters
in the constellation of Taurus,
between Orion and Auriga.
400 ly away.

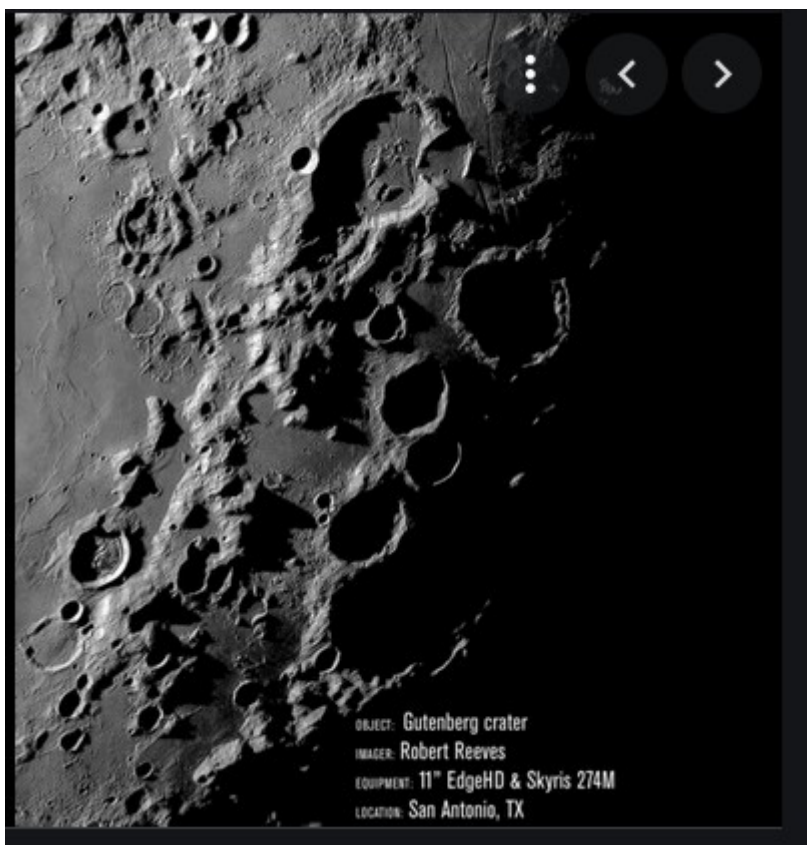


cruse along the milky way see the tail of the scorpion glowing cloud of gas



<https://earthsky.org/brightest-stars/scorpion-stinger-stars-also-called-cats-eyes/>





TELESCOPE

Constellation of Cygnus and check the end star its a double star.

Albireo. (gold and blue)

Abireo as one of the most beautiful binary systems in the heavens



Albireo is about 430 light-years

Separated by 35 seconds of arc

their orbital period is probably at least 100,000 years.

<https://www.syfy.com/syfy-wire/long-standing-astronomical-mystery-solved-albireo-is-not-a-binary-star>

South

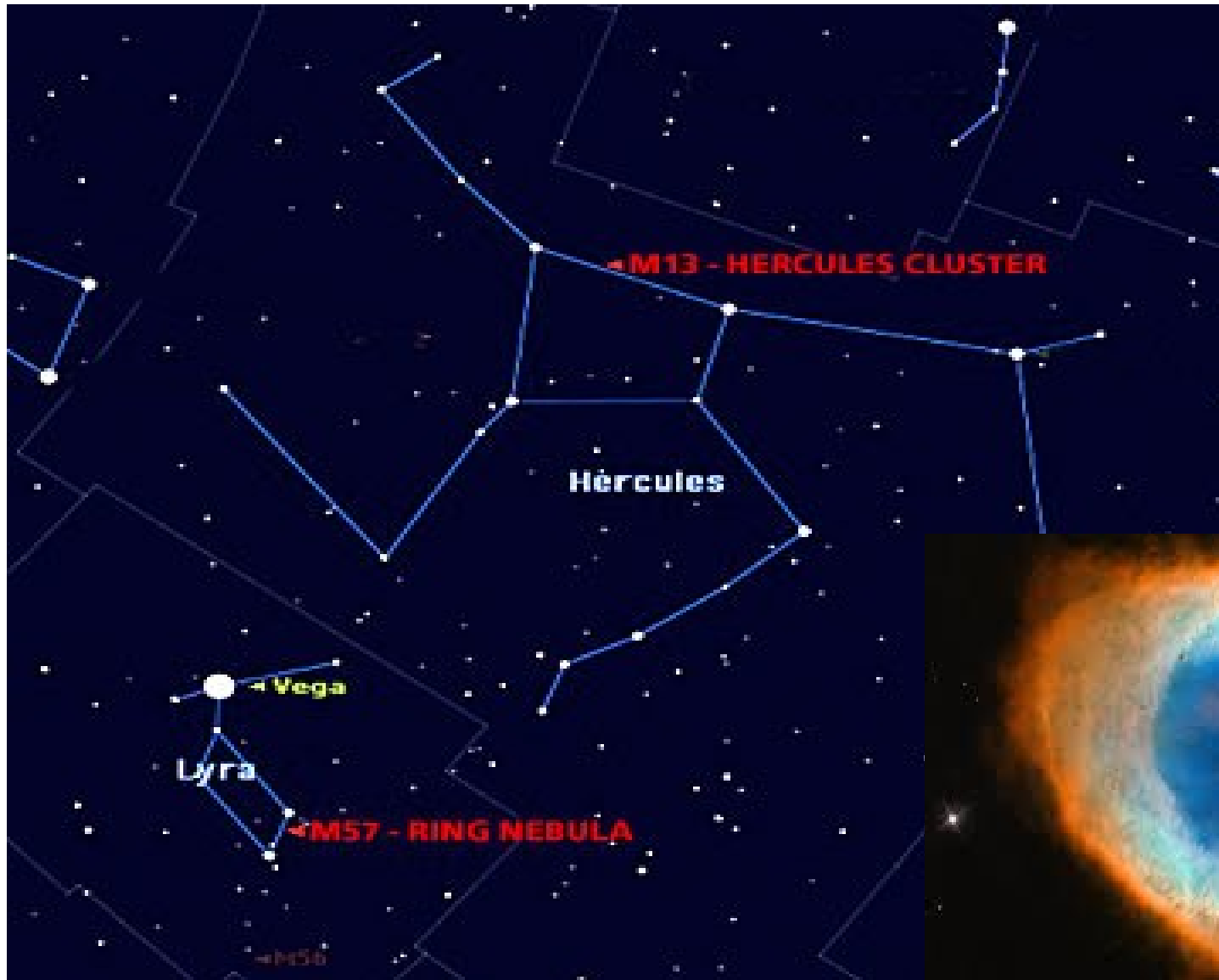


In Hercules next to Cygnus : great globular cluster in
Hercules
(in the head)

200 galaxies[3] some 500 million light-years



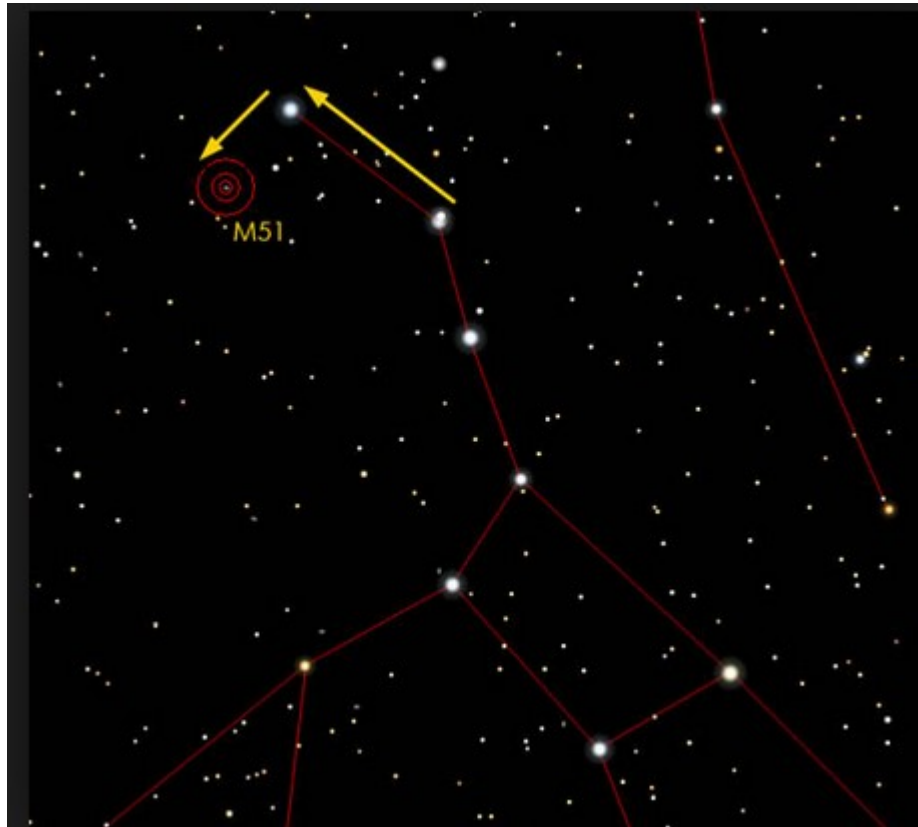
go back to Cygnus (northern cross) go to Lyra and you can see the ring nebula ejected by a dying star. cosmic burp.



(2,300 light-years)



end of big dipper you find the whirlpool galaxy. Galaxy 20 million light years away.



winter



M=-1.5 down left of belt
Canis major, the dog star
It is a binary system. Sirius A and B
Sirius A is the brighter. They are both
White dwarf.

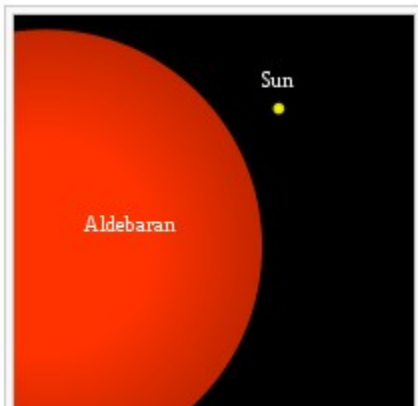


winter



65 light years
Easiest to see

M = 0.8 extend the belt up
in Taurus the Bull
New planet found in orbit.
RED GIANT (see pic)
Binary system Its is a red giant.
Brighter than the sun because bigger.



winter



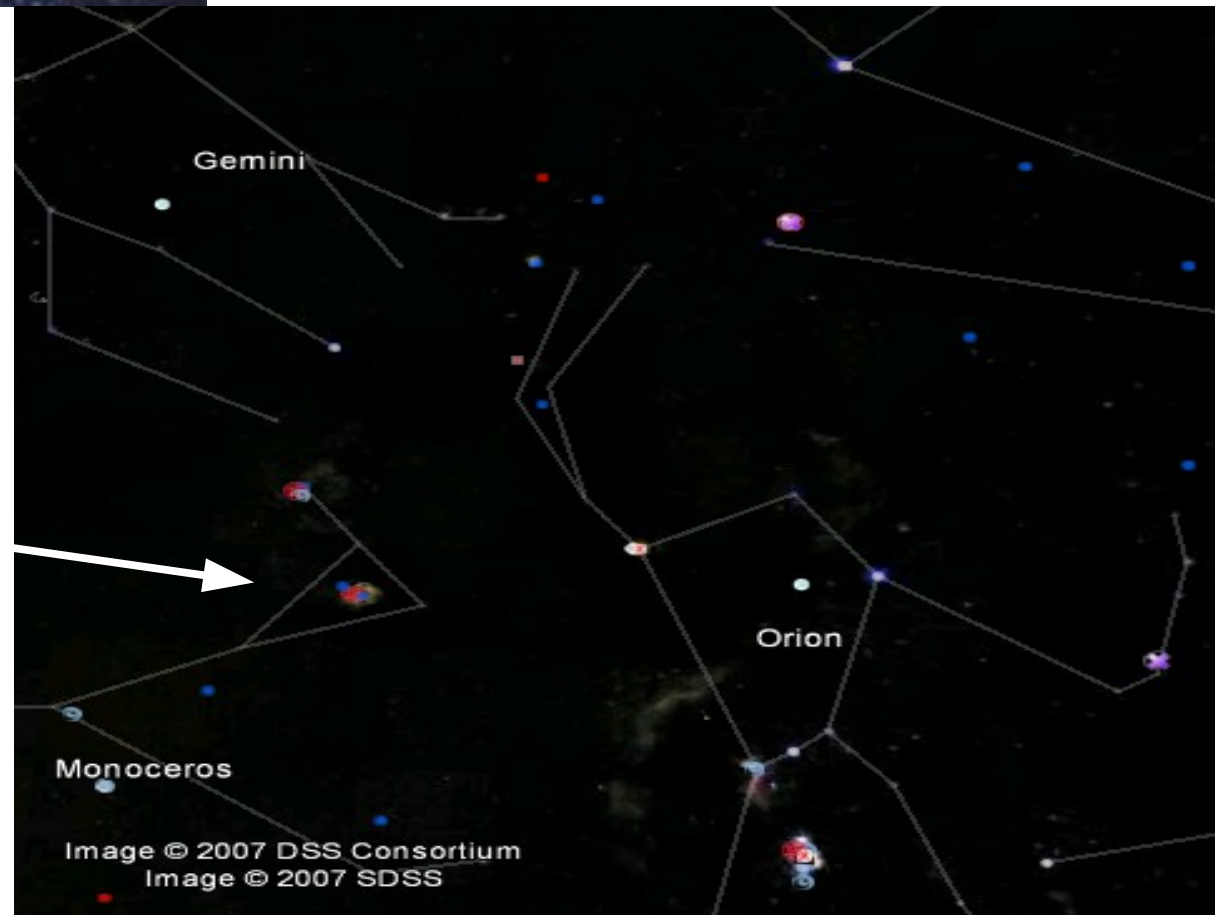
M=3.8 left to Orion in Canis Minor

The little dog star.

Twice as big as the sun.

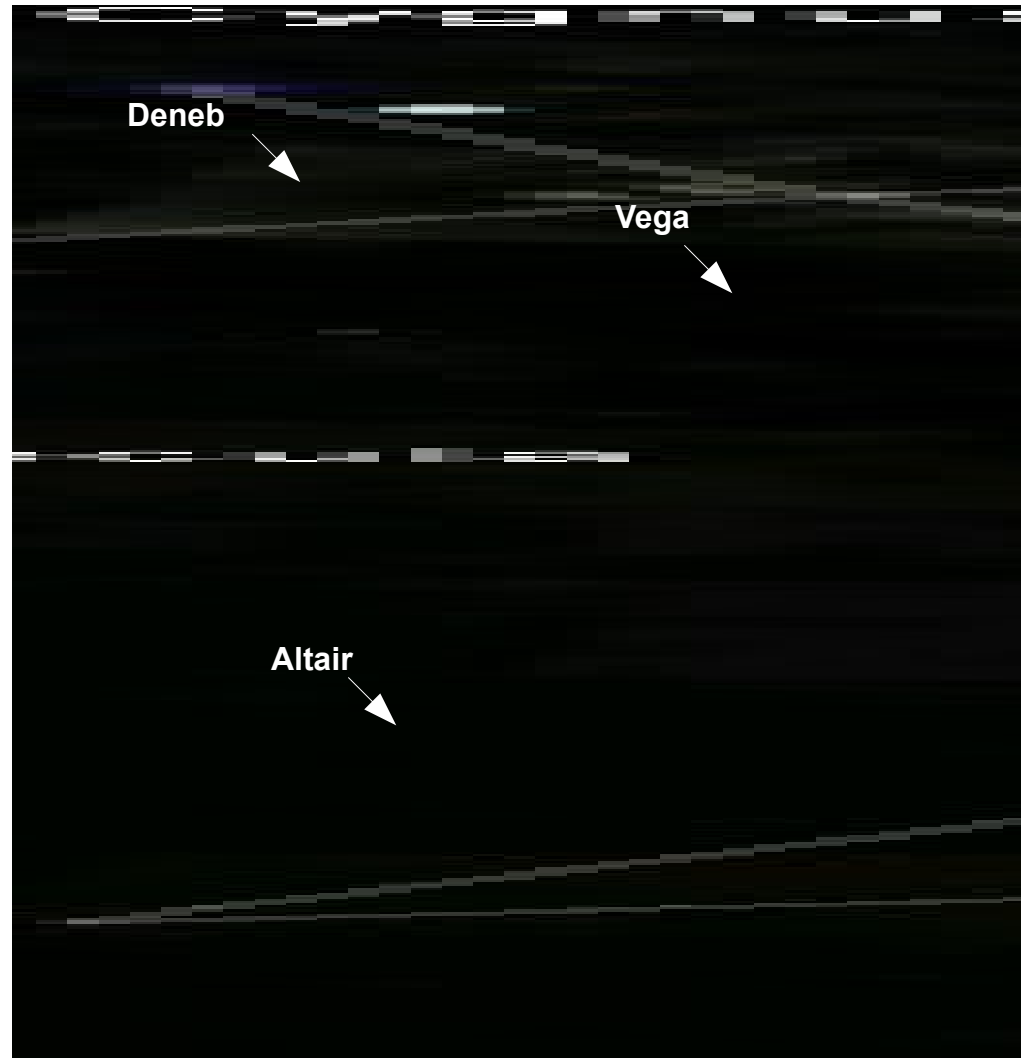
Yellow-white in color. Hotter than the sun

Will end as a red giant but bigger than the sun/



Summer in summer triangle / EAGLE

Vega (constellation Lyra) Altair .77 (in aquila the eagle)
DENEK 1.25 IN SWYNE (cygnus) see the head.



Altair is a blue-white star. Multiple system
Vega is blue -white
Deneb is a blue super giant. Is expected to explode
In a few million years. Reached end of life.